Graphs and Transformations

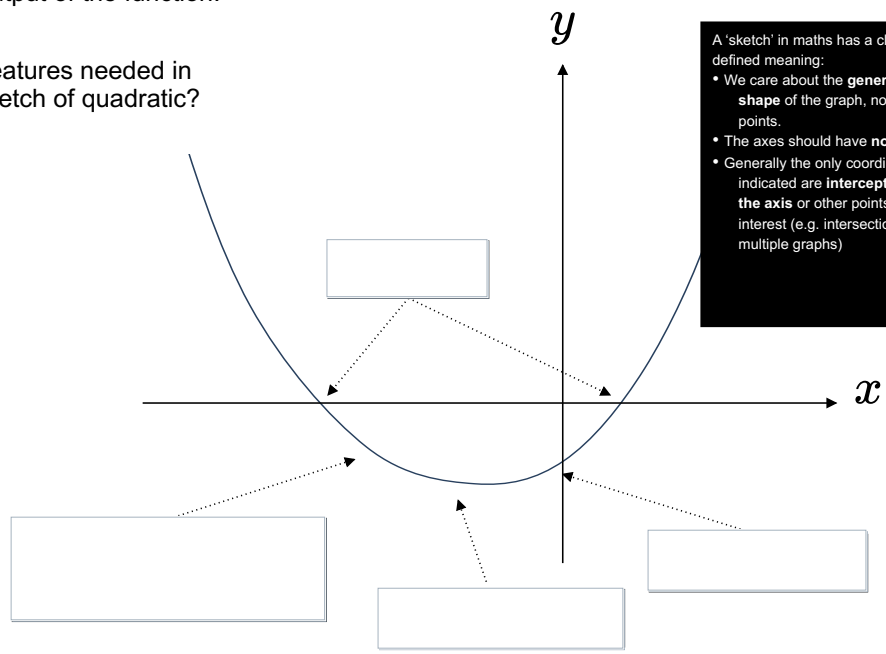
Chapter 2 · Original slide 26

4: Quadratic Graphs

26

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

Features needed in sketch of quadratic?



A 'sketch' in maths has a clearly defined meaning:

- We care about the **general shape** of the graph, not exact points.
- The axes should have **no scale**.
- Generally the only coordinates indicated are **intercepts with the axis** or other points of interest (e.g. intersections of multiple graphs)

Chapter 2 · Original slide 27

Example

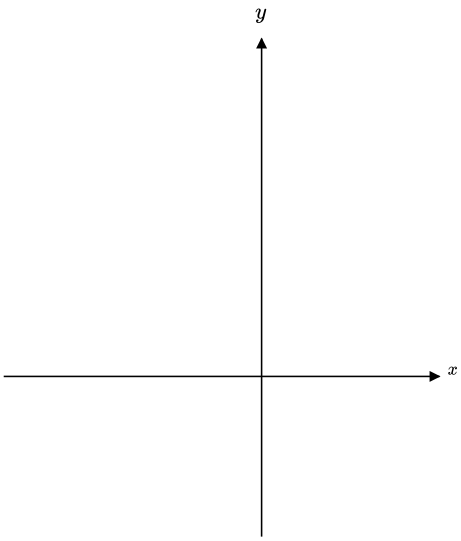
27

Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

y-intercept:

Turning point:



Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

Graphs and Transformations

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Example

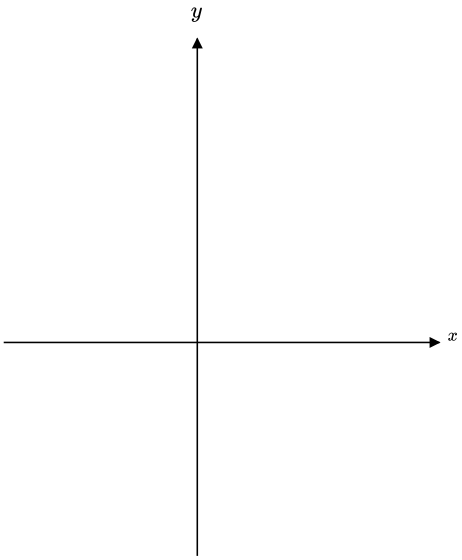
28

Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

Roots:

y-intercept:

Turning point:



Line of reflection:

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Test Your Understanding

29

Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a $y = x^2 + 4$

b $y = x^2 - 7x + 10$

c $y = 5x + 3 - 2x^2$

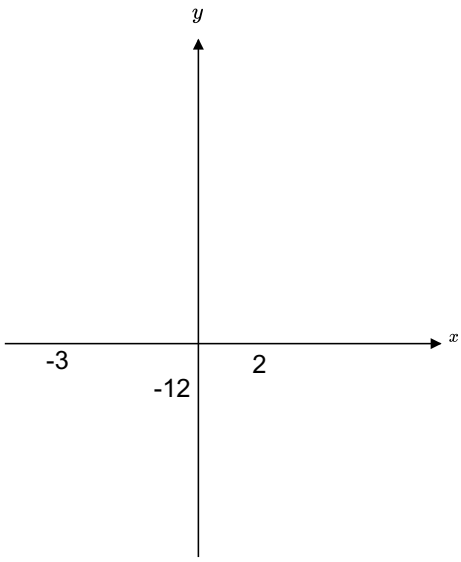
d $y = x^2 + 4x + 11$

Graphs and Transformations

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Determining the Equation using a Graph

30

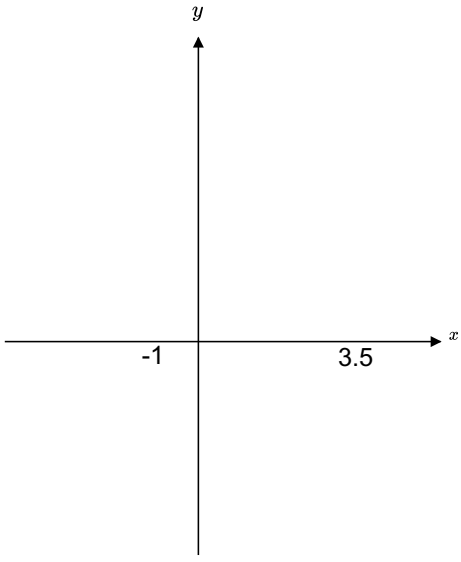


Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

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Further Example

31



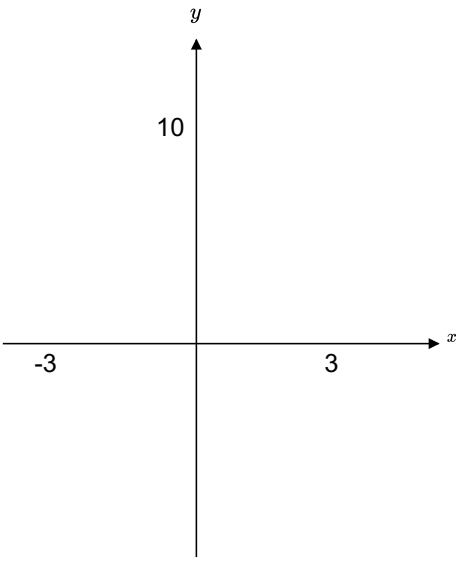
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

Graphs and Transformations

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Test Your Understanding

32



Determine an equation of this quadratic graph.

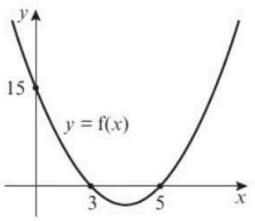
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Exercise 2F Pure Y1 Page 30

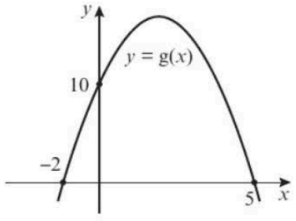
33

2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a, b, c for each function.

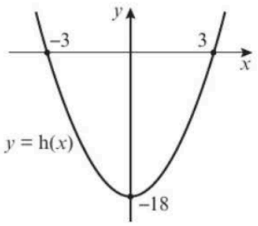
a



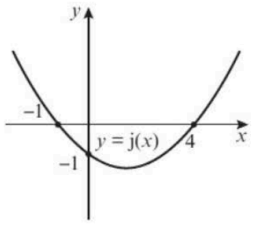
b



c



d



Graphs and Transformations

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Exercise 2F Pure Y1 Page 30

34

3 The graph of $y = ax^2 + bx + c$ has a minimum at $(5, -3)$ and passes through $(4, 0)$.
Find the values of a, b, c.

Extension Question:
[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?
$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

Chapter 4 · Original slide 4

Polynomial Graphs

4

In Chapter 2 we briefly saw that a **polynomial** expression is of the form:

$$a + bx + cx^2 + dx^3 + ex^3 + \dots$$

where $a, b, c, d, e, \dots$ are constants (which could be 0).

The **order** of a polynomial is its highest power.

Order	Name
0	Constant (e.g. “4”)
1	
2	
3	
4	
5	

These are covered in Chapter 5.

Chapter 2 explored the graphs for these.

We will cover these now.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

Graphs and Transformations

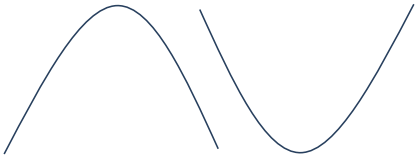
Chapter 4 · Original slide 5

Polynomial Graphs

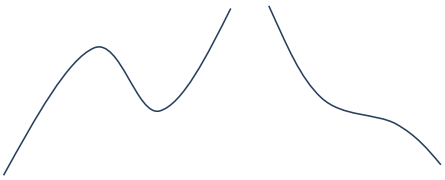
5

Order:

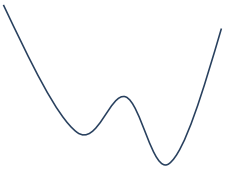
2



3



4



What property connects the order of the polynomial and the shape?

In Chapter 2 how did we tell what way up a quadratic is, and why does this work?

Chapter 4 · Original slide 6

Polynomial Graphs

6

e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive y value. Thus as $x \to \infty$, $y \to \infty$

Equation	If $a > 0$	Resulting Shape	If $a < 0$	Resulting Shape
$y = ax^2 + bx + c$	As $x \to \infty, y \to \infty$ As $x \to -\infty, y \to \infty$		As $x \to \infty, y \to -\infty$ As $x \to -\infty, y \to -\infty$	
$y = ax^3 + bx^2 + cx + d$				
$y = ax^4 + bx^3 + cx^2 + dx + e$				
$y = ax^5 + bx^4 + \ldots$				

If $a > 0$, what therefore can we say about the shape if:
- **The order is odd**: It goes
- **The order is even**: The ta

Graphs and Transformations

Chapter 4 · Original slide 7

Cubics

7

Sketch the curve with equation
$y = (x-2)(1-x)(1+x)$

Features you must consider:

Shape?

Roots?

y-intercept?

Sketch the curve with equation
$y = x^2(x-1)$

Shape?

Roots?

y-intercept?

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Cubics

8

Sketch the curve with equation
$y = (2-x)(x+1)^2$

Shape?

Roots?

y-intercept?

Sketch the curve with equation
$y = (x-4)^3$

Shape?

Roots?

y-intercept?

Graphs and Transformations

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Cubics with Limited Roots

9

Sketch the curve with equation

$$y = (x + 1)\left(x^2 + x + 1\right)$$

Shape?

Roots?

y-intercept?

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Finding the equation yourself

10

Edexcel C1 May 2013(R) Q9

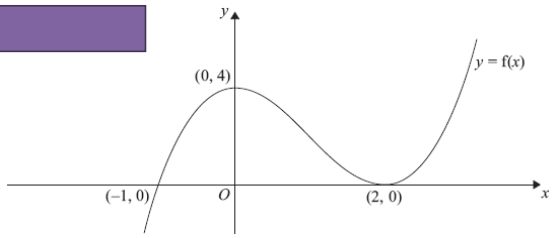


Figure 1 shows a sketch of the curve C with equation $y = f(x)$.
The curve C passes through the point (–1, 0) and touches the x-axis at the point (2, 0).
The curve C has a maximum at the point (0, 4).
The equation of the curve C can be written in the form.

$$y = x^3 + ax^2 + bx + c$$

where a, b and c are integers.
(a) Calculate the values of a, b, c.

Graphs and Transformations

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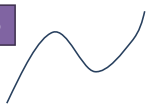
Test Your Understanding

11

1 Sketch the curve with equation $y = x(x-3)^2$

2 Sketch the curve with equation $y = -(x+2)^3$

3



A curve has this shape, touches the x axis at 3 and crosses the x axis at -2. Give a suitable equation for this graph.

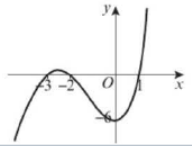
Sketch the curve with equation $y = 2x^2(x-1)(x+1)^3$

Chapter 4 · Original slide 12

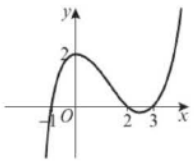
Exercise 4A –Y1 Pure pages 63

12

5 The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b, c, d are real constants.
a Find the values of b, c, d. (3 marks)
b Write down the coordinates of the point where the curve crosses the y-axis. (1 mark)



6 The graph of $y = ax^3 + bx^2 + cx + d$ is shown opposite, where a, b, c, d are real constants. Find the values of a, b, c, d. (4 marks)



Graphs and Transformations

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Exercise 4A –Y1 Pure pages 63

13

7 Given that $f(x) = (x-10)(x^2-2x)+12x$

a Express $f(x)$ in the form $x(ax^2+bx+c)$ where a, b, c are real constants.

b Hence factorise $f(x)$ completely.

c Sketch the graph of $y = f(x)$ showing clearly the points where the graph intersects the axes.

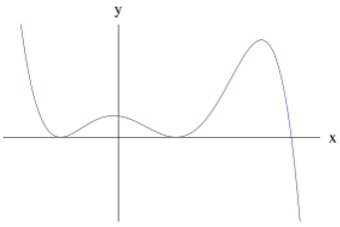
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Exercise 4A Extension

14

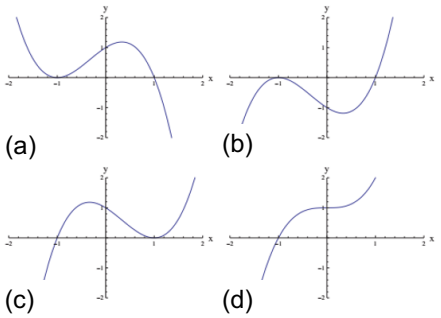
Extension

1 [MAT 2012 1E] Which one of the following equations could possibly have the graph given below?



A) $y = (3-x)^2(3+x)^2(1-x)$
B) $y = -x^2(x-9)(x^2-3)$
C) $y = (x-6)(x-2)^2(x+2)^2$
D) $y = (x^2-1)^2(3-x)$

2 [MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?



Graphs and Transformations

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Recap

15

If we sketched $y = (x - a)(x - b)^2(x - c)^3$ what happens on the x-axis at:

$x = a$:

$x = b$:

$x = c$:

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Quartics

16

If you understand the principle of sketching polynomials in general, then sketching quartics shouldn't feel like anything new.
Recall that if the x^4 term is positive, the 'tails' both go upwards, otherwise downwards.

Sketch the curve with equation
$y = x(x + 1)(x - 2)(x - 3)$

Shape:
Roots:
y-intercept:

Sketch the curve with equation
$y = (x - 2)^2(x + 1)(3 - x)$

Shape:
Roots:
y-intercept:

Graphs and Transformations

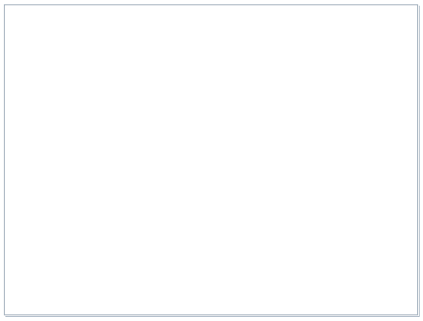
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Quartics

17

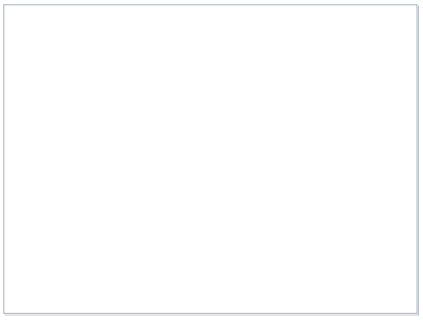
Sketch the curve with equation
$y = (x+1)(x-1)^3$

-1 root only appears once so line crosses at $x = -1$
+1 root triple repeated so point of inflection at $x = 1$



Sketch the curve with equation
$y = (x-2)^4$

2 is a quadruple repeated root! Because the line effectively crosses the axis 4 times all at -2, it ends up in the opposite direction, and hence looks like a ‘touch’ point.

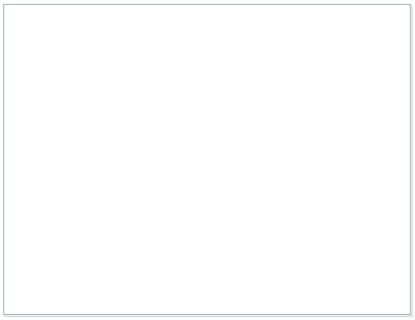


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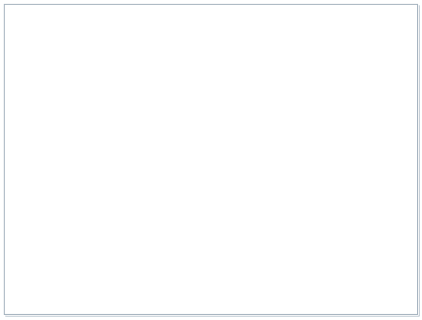
Test Your Understanding

18

Sketch the curve with equation
$y = x^2(x+1)(x-1)$



Sketch the curve with equation
$y = -(x+1)(x-3)^3$



Graphs and Transformations

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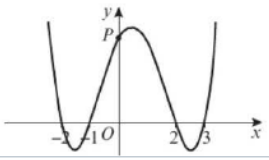
Exercise 4B – Pure Y1 Pages 65-66

19

3 The graph of $y = x^4 + bx^3 + cx^2 + dx + e$ is shown opposite, where b, c, d, e are real constants.

a Find the coordinates of point P. (2 marks)

b Find the values of b, c, d, e. (3 marks)



4 Sketch the graph of $y = (x + 5)(x - 4)(x^2 + 5x + 14)$.

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Exercise 4B Extension

20

Extension

1 *[STEP I 2012 Q2a]*

a. Sketch $y = x^4 - 6x^2 + 9$

b. For what values of b does the equation $y = x^4 - 6x^2 + b$ have the following number of distinct roots (i) 0, (ii) 1, (iii) 2, (iv) 3, (v) 4.

a)

b) By changing b, we shift the graph up and down. Then we can see that:

i) 0 $b > 0$

ii) 1

iii)

iv)

v)

vi)

vii)

Graphs and Transformations

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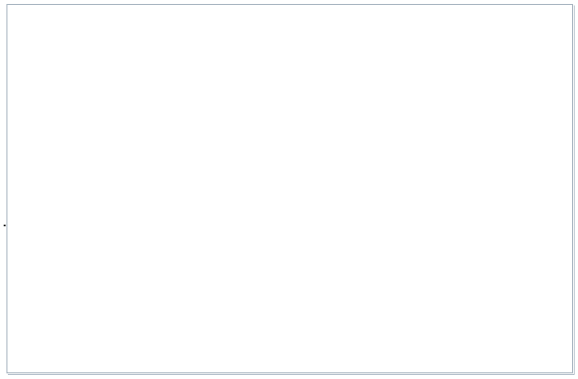
GCSE RECAP: Reciprocal Graphs

21

Sketch $y = \frac{1}{x}$



Sketch $y = -\frac{3}{x}$





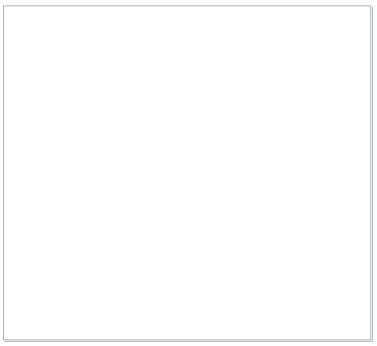
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Reciprocal Graphs

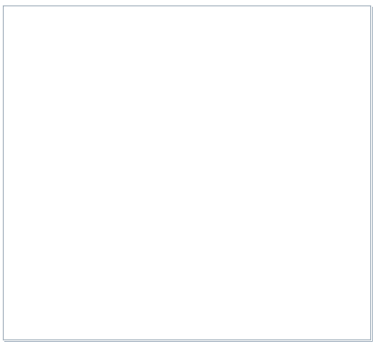
22

Sketch $y = \frac{3}{x^2}$

This is new to the A Level 2017 syllabus.



Sketch $y = -\frac{4}{x^2}$



Hint: Note that anything squared will always be at least 0.

Graphs and Transformations

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Reciprocal Graphs

23

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

Chapter 4 · Original slide 24

Exercise 4C – Pure Y1 page 67

24

1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$

d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

Graphs and Transformations

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Exercise 4C – Pure Y1 page 67

25

2 Use a separate diagram to sketch each pair of graphs. **a** $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$

Chapter 4 · Original slide 34

Transformations of Functions

34

Suppose $f(x) = x^2$

Sketch $y = f(x)$:

Then $f(x + 2) =$

Sketch $y = f(x + 2)$

What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x + 2)$?

Graphs and Transformations

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Transformations of Functions

35

This is all you need to remember when considering how transforming your function transforms your graph...

	Affects which axis?	What we expect or opposite?
Change **inside** $f()$		
Change **outside** $f()$		

Therefore...

$y = f(x-3)$ →

$y = f(x) + 4$ →

$y = f(5x)$ →

$y = 2f(x)$ →

Chapter 4 · Original slide 36

Sketching transformed graphs

36

Sketch $y = x^2 + 3$

Sketch $y = \frac{2}{x+1}$

Graphs and Transformations

Chapter 4 · Original slide 37

More Examples

37

Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$$y = (2x)^2(2x-4)$$

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Reflections of Graphs

38

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

Graphs and Transformations

Chapter 4 · Original slide 39

Test Your Understanding

39

If $y = (x+1)(x-2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.

Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.

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Exercise 4E – Pure Y1 page 74-75 (translations)

40

7 The point $P(4, -1)$ lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x-2)$.

b State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x) + 3$.

f
o

8 The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

Graphs and Transformations

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Exercise 4E – Pure Y1 page 74-75 (translations)

41

11 a Sketch the graph of $y = x^3 + 4x^2 + 4x$. **b** The poin
coordinates $(-1, 0)$ lies on the curve with equation
$y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find
the two possible values of a.

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Exercise 4E – Pure Y1 page 74-75 (translations)

42

12 a Sketch the graph of $y = x(x + 1)(x + 3)^2$. **b** Find the possible values of b such that
the point $(2, 0)$ lies on the curve with equation $y = (x + b)(x + b + 1)(x + b + 3)^2$.

b
c
o
u

e

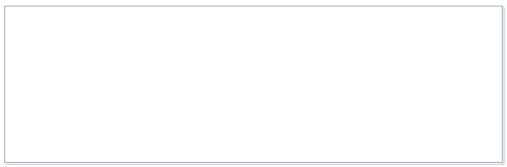
Graphs and Transformations

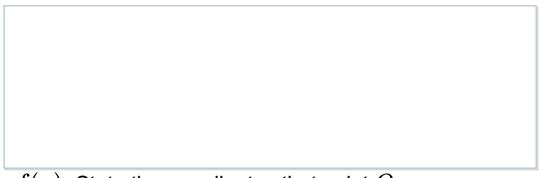
Chapter 4 · Original slide 43

Exercise 4F – Pure 1 Page 78 (stretches/reflections)

43

7 The point $P(2, -3)$ lies on the curve with equation $y = f(x)$. **a** State the coordinates that point P is transformed to on the curve with equation $y = f(2x)$. **b** State the coordinates that point P is transformed to on the curve with equation $y = 4f(x)$.





8 The point $Q(-2, 8)$ lies on the curve with equation $y = \mathrm{f}(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = \mathrm{f}(\frac{1}{2}x)$.

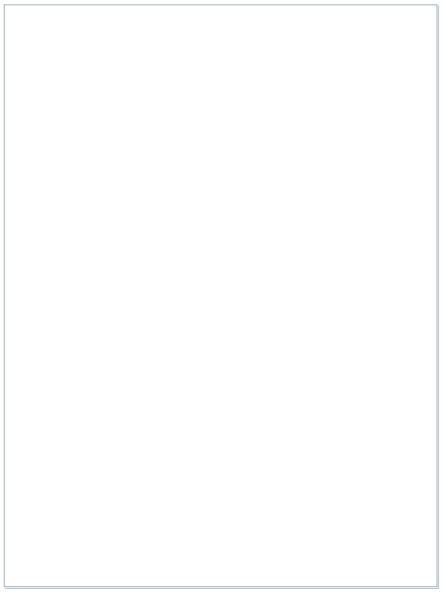


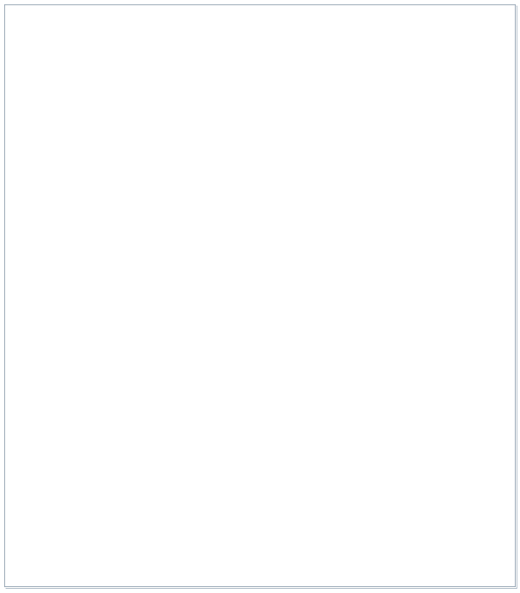
Chapter 4 · Original slide 44

Exercise 4F – Pure 1 Page 78 (stretches/reflections)

44

9 a Sketch the graph of $y = (x - 2)(x - 3)^2$. **b** The graph of $y = (ax - 2)(ax - 3)^2$ passes through the point $(1, 0)$. Find two possible values for a.





Graphs and Transformations

Chapter 4 · Original slide 45

Effect of transformation on specific points

45

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform.
Where would each of these points end up?

$\mathbf{y=}f(x)$	$(4,3)$	$(1,0)$	$(6,-4)$
$y=f(x+1)$			
$y=f(2x)$			
$y=3f(x)$			
$y=f(x)-1$			
$y=f\left(\frac{x}{4}\right)$			
$y=f(-x)$			
$y=-f(x)$			

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Test Your Understanding

46

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Q10

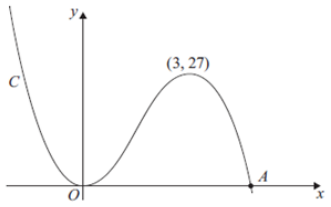


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = f(x)$, where $f(x) = x^2(9 - 2x)$. There is a minimum at the origin, a maximum at the point $(3, 27)$ and C cuts the x-axis at the point A.

(a) Write down the coordinates of the point A. (1)

(b) On separate diagrams sketch the curve with equation

(i) $y = f(x + 3)$,

(ii) $y = f(3x)$.

On each sketch you should indicate clearly the coordinates of the maximum point and any points where the curve cross or meet the coordinate axes. (3)

The curve with equation $y = f(x) + k$, where k is a constant, has a maximum point at $(3, 10)$.

(c) Write down the value of k. (1)

(a)

(b)(i)

(ii)

(c)

Graphs and Transformations

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Exercise 4G Pure Y1 Pages 80

47

5 The point $P(2, 1)$ lies on the graph with equation $y = \mathrm{f}(x)$.

a On the graph of $y = \mathrm{f}(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a.

b Write down the coordinates of the point to which P maps under each transformation

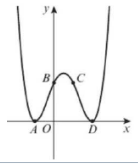
i $\mathrm{f}(x - 4)$ ii $3\mathrm{f}(x)$ iii $\frac{1}{2}\mathrm{f}(x) - 4$

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Exercise 4G Pure Y1 Pages 80

48

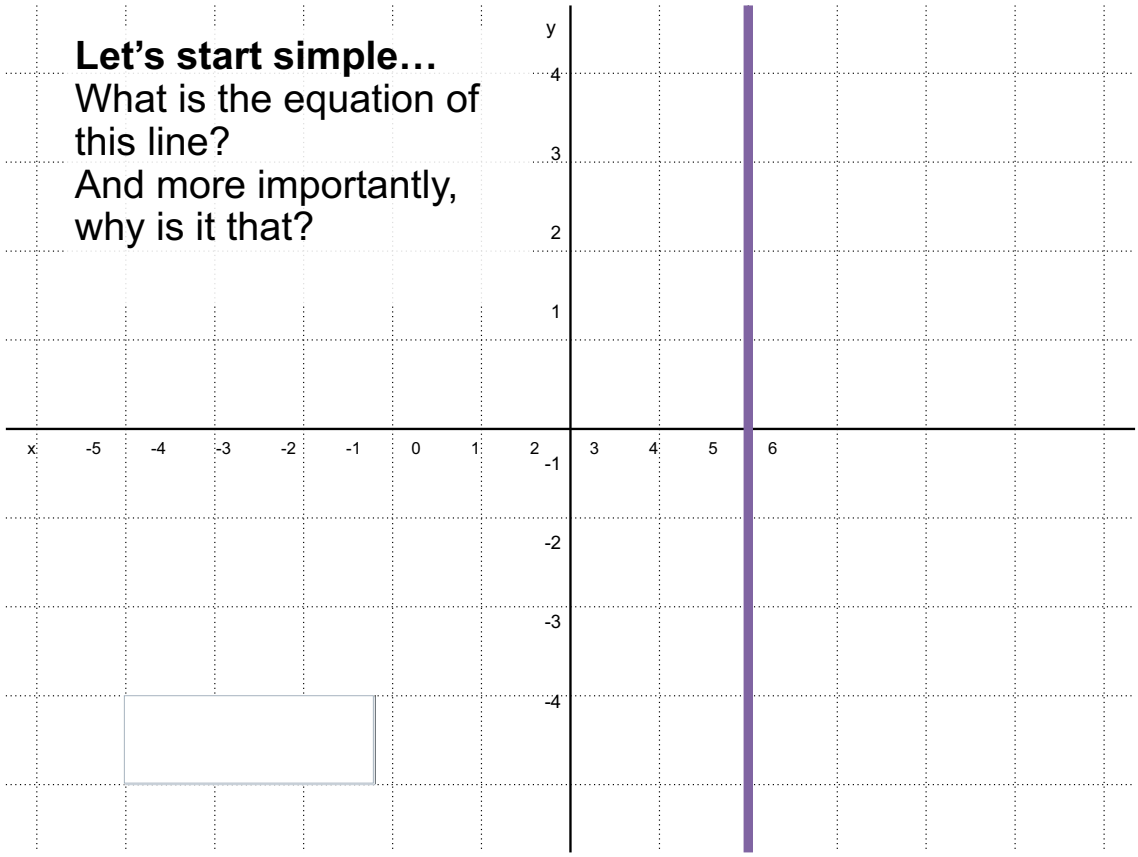
6 The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points $A(-1, 0), B(0, 2), C(1, 2), D(2, 0)$ lie on the curve. Sketch the following graphs and give the coordinates of the points A, B, C, D after each transformation:



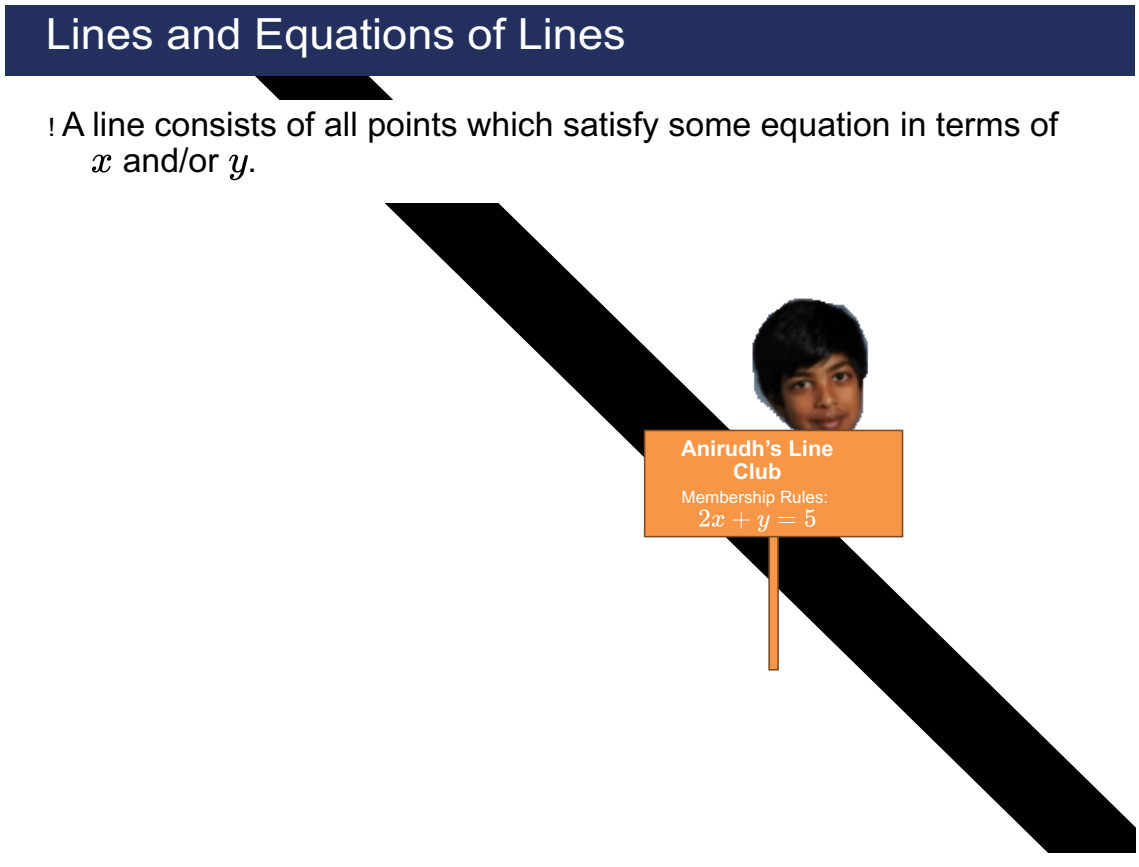
a $y + 2 = \mathrm{f}(x)$ **b** $\frac{1}{2}y = \mathrm{f}(x)$ **c** $y - 3 = \mathrm{f}(x)$ **d** $3y = \mathrm{f}(x)$ **e** $2y - 1 = \mathrm{f}(x)$

Graphs and Transformations

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Chapter 5 · Original slide 5



Graphs and Transformations

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Examples

This means we can **substitute** the values of a coordinate into our equation whenever we know the point lies on the line.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

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Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

a) y-axis:

b) x-axis:

What mistakes do you think it's easy to make?

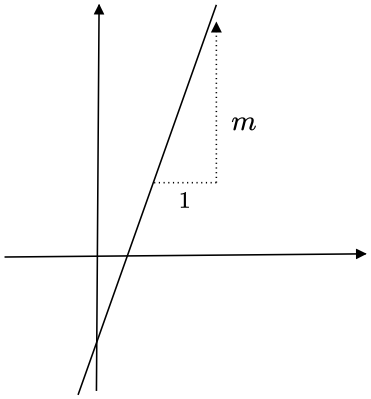
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- **S**

Graphs and Transformations

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Recap of gradient

The steepness of a line is known as the **gradient**.
It tells us what y changes by as x increases by 1.



So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?
How would that give us a suitable formula for the gradient m?

$m =$ ☐

Chapter 5 · Original slide 9

Examples

Find the gradient of the line that goes through the points:

1 $(1, 4)$ $(3, 10)$ $m =$ ☐

2 $(5, 7)$ $(8, 1)$ $m =$ ☐

3 $(2, 2)$ $(-1, 10)$ $m =$ ☐

4 Show that the points
$\mathrm{A}\,(3, 4), B\,(5, 5), C\,(11, 8)$ all lie on a straight

Graphs and Transformations

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Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1. Work out the value of a.

$\frac{a}{}$

$\frac{a}{}$

a

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$y = mx + c$

One form we can put a straight line equation in is:

$$y = mx + c$$

Gradient

y-intercept

Why does it work?

Graphs and Transformations

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$y = mx + c$

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

$3y$

$y =$

Make y the subject so we have the form $y = mx + c$
Put y on the side it's positive.

Chapter 5 · Original slide 13

$ax + by + c = 0$

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the '**slope-intercept form**'.
But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers.
This is known as the '**standard**' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$…

Graphs and Transformations

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Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$y = mx + c$
"Slope-Intercept Form"

$ax + by + c = 0$
"Standard Form"

Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

$x + 4 = 0$

Symmetry

In general, the '**linear combination**' of two variables x and y is $ax + by$, i.e. "some amount of x and some amount of y". There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the '**dot product**' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

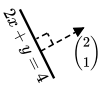
thus since $ax + by + c = 0$, we can represent a straight line using:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 \qquad (1)$$

We can extend to 3D points to get the equation of a **plane**:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 \qquad (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is **perpendicular to the line**. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$2x + y = 4$ ➡ $\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$ ➡ 

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Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

$5y$
$2x$

Graphs and Transformations

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Exercise 5A – Pure Y1 Page 91

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. $(3\ \mathrm{ma}\ldots$

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Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

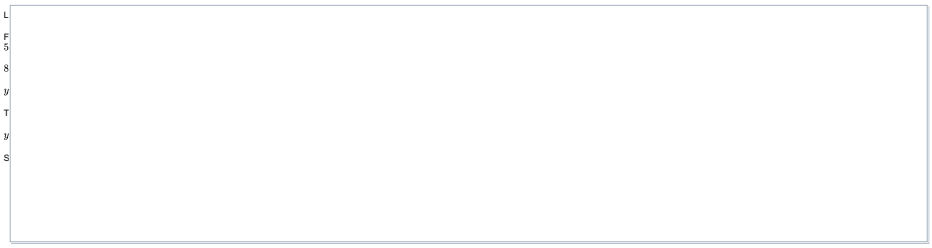
Find the values of a and c. (2 marks)

Graphs and Transformations

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Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.



12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.

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Sin Graph

44

What does it look like?

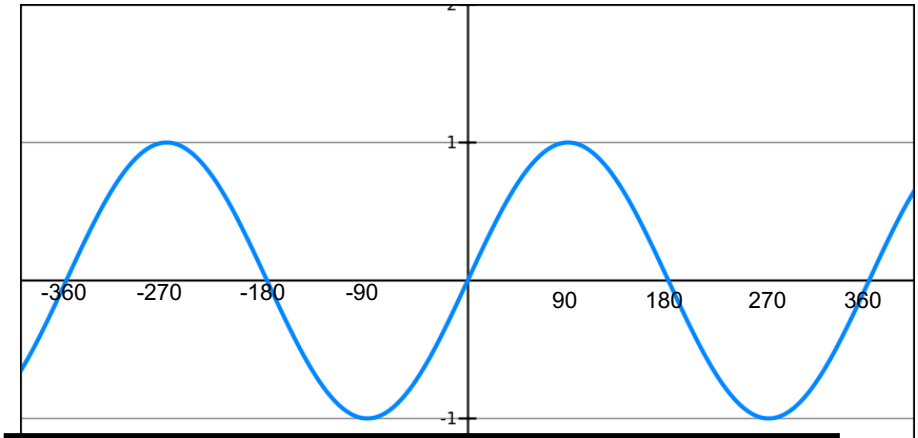
Graphs and Transformations

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Sin Graph

45

What do the following graphs look like?



Suppose we know that **sin(30) = 0.5**. By thinking about symmetry in the graph, how could we work out:

sin(150) = sin(-30) = sin(210) =

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Cos Graph

46

What do the following graphs look like?

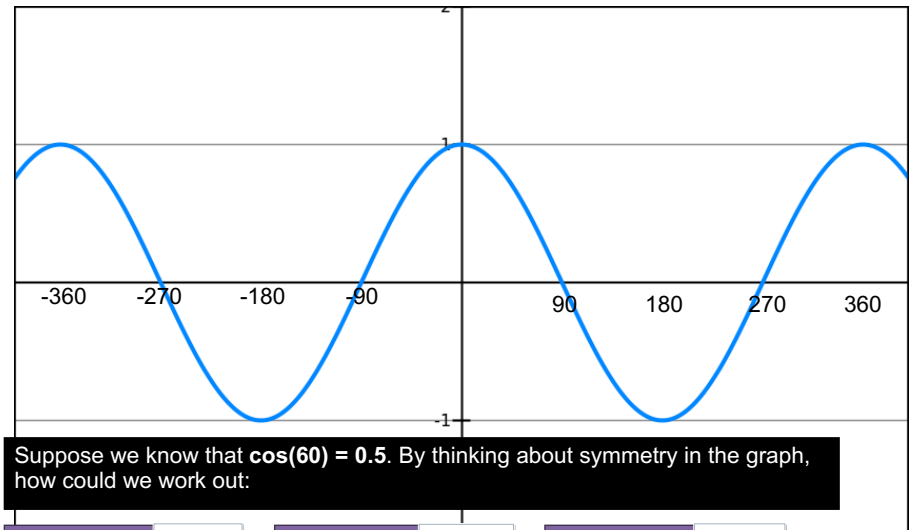
Graphs and Transformations

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Cos Graph

47

What does it look like?



Suppose we know that **cos(60) = 0.5**. By thinking about symmetry in the graph, how could we work out:

cos(120) = -0.5 | cos(-60) | cos(240) = -0.5

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Tan Graph

48

What does it look like?

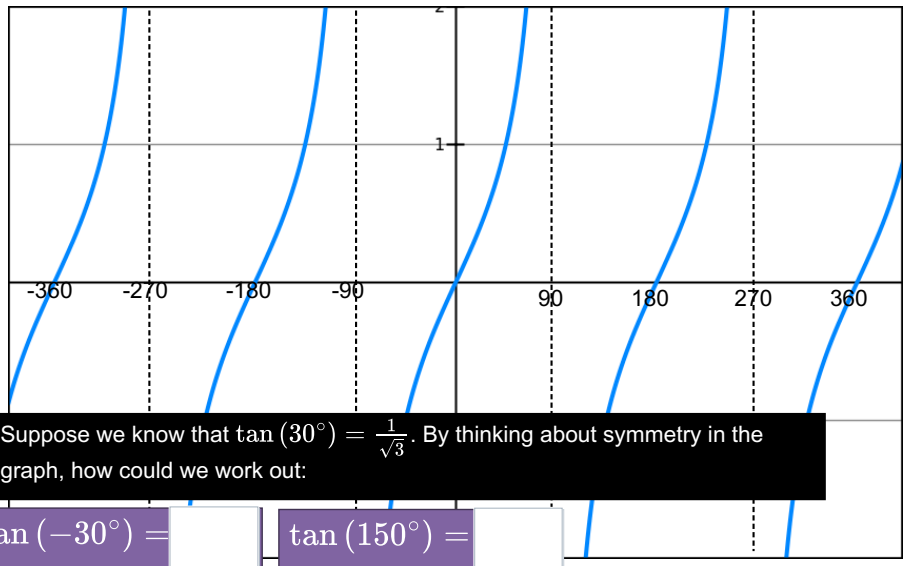
Graphs and Transformations

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Tan Graph

49

What does it look like?



Suppose we know that $\tan(30^\circ) = \frac{1}{\sqrt{3}}$. By thinking about symmetry in the graph, how could we work out:

$\tan(-30^\circ) =$ $\tan(150^\circ) =$

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Transforming Trigonometric Graphs

50

There is no new theory here: just use your knowledge of transforming graphs, i.e. whether the transformation occurs 'inside' the function (i.e. input modified) or 'outside' the function (i.e. output modified).

Sketch $y = 4\sin x$, $0 \leq x \leq 360^\circ$

Sketch $y = \cos(x + 45^\circ)$, $0 \leq x \leq 360^\circ$

Graphs and Transformations

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Transforming Trigonometric Graphs

51

Sketch $y = -\tan x,\ 0 \leq x \leq 360^\circ$

Sketch $y = \sin\left(\frac{x}{2}\right),\ 0 \leq x \leq 360^\circ$

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Exercise 9F – Y1 Page 194

52

1 Sketch the graph of $y = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par2 Sketch the graph of $y = \tan\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.\par3 Sketch the graph of $y = \sin\theta$ in the interval $-90^\circ \leq \theta \leq 270^\circ$.

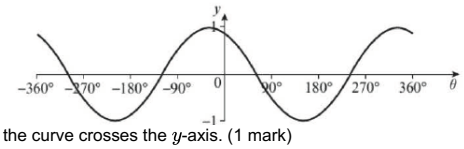
Graphs and Transformations

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Exercise 9G – Y1 Pages 197-198

53

7 The graph shows the curve
$y = \cos(x + 30^\circ), -360^\circ \leq x \leq 360^\circ$.
a Write down the coordinates of the points where the curve crosses the x-axis. (2 marks)



b Find the coordinates of the point where the curve crosses the y-axis. (1 mark)

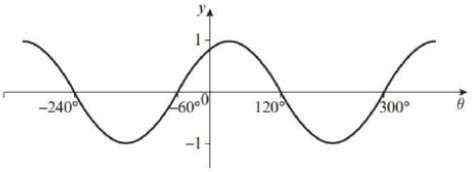
b $\cos 30^\circ = \frac{\sqrt{3}}{2}$; $\left(0, \frac{\sqrt{3}}{2}\right)$

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Exercise 9G – Y1 Pages 197-198

54

8 The graph shows the curve with equation $y = \sin(x + k), -360^\circ \leq x \leq 360^\circ$, where k is a constant.
a Find one possible value for k. (2 marks)
b Is there more than one possible answer to part a? Give a reason for your answer. (2 marks)



Graphs and Transformations

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Exercise 9G – Y1 Pages 197-198

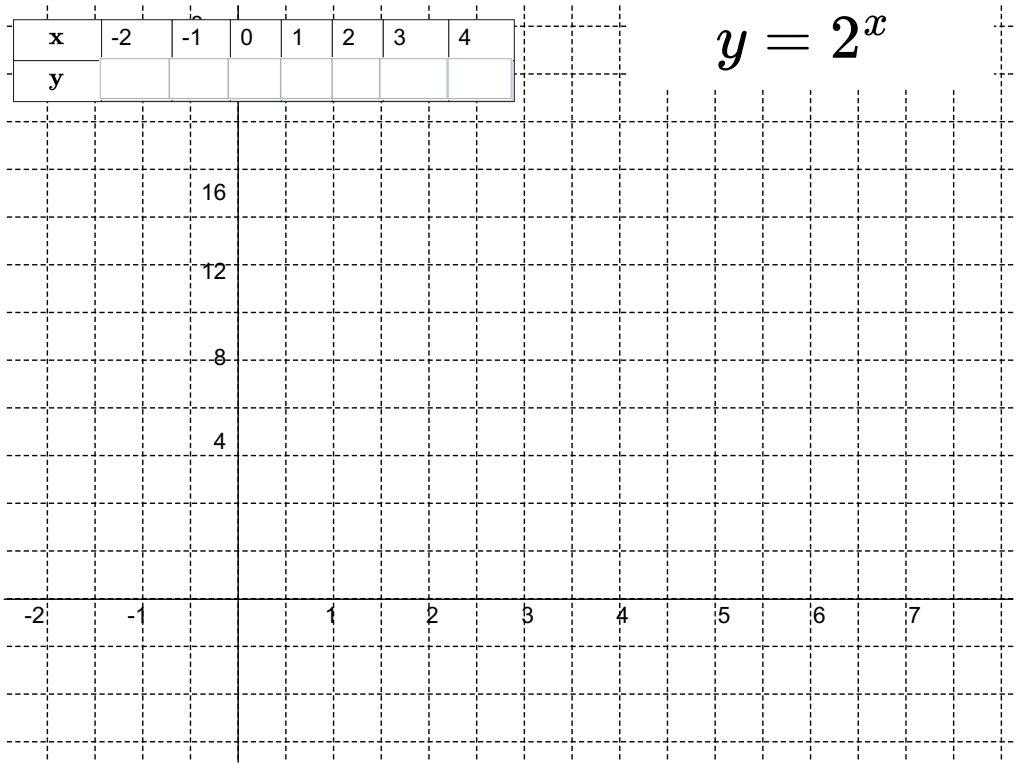
55

9 The variation in the depth of water in a rock pool can be modelled using the function $y = \sin(30t)^\circ$, where t is the time in hours and $0 \leq t \leq 6$.
a Sketch the function for the given interval.
b If $t = 0$ represents midday, during what times will the rock pool be at least half full?

Chapter 14 · Original slide 4

x	-2	-1	0	1	2	3	4
y							

$y = 2^x$



Graphs and Transformations

Chapter 14 · Original slide 5

Contrasting exponential graphs

5

On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$

On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

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Graph Transformations

6

Sketch $y = 2^{x+3}$

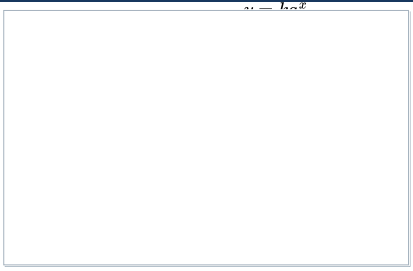
Graphs and Transformations

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Exercise 14A (Pages 313-314)

7

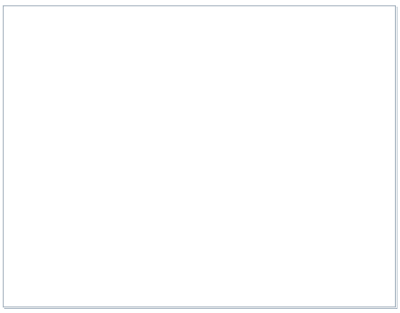
6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

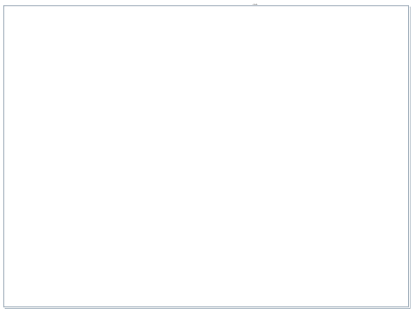


7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.



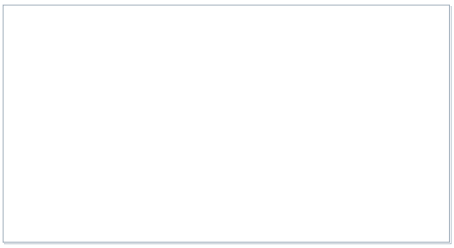


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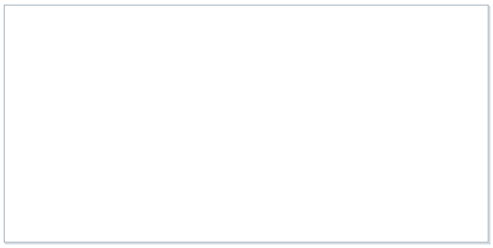
More Graph Transformations

11

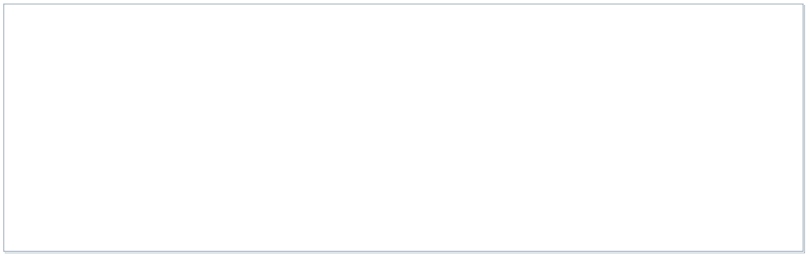
Sketch $y = e^{3x}$



Sketch $y = 5e^{-x}$



Sketch $y = 2 + e^{\frac{1}{3}x}$



Graphs and Transformations

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Test Your Understanding

12

Sketch $y = e^{-2x} - 1$

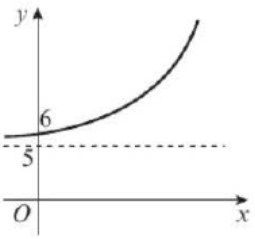
Chapter 14 · Original slide 13

Exercise 14B (Pages 316-317)

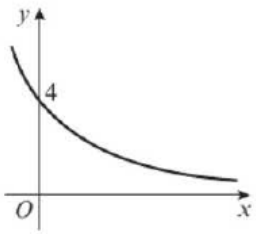
13

4 Each of the sketch graphs below is of the form $y = Ae^{bx} + C$, where A, b and C are constants. Find the values of A and C for each graph, and state whether b is positive or negative.

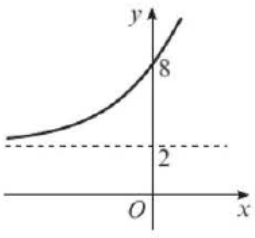
a



b



c



Graphs and Transformations

Chapter 14 · Original slide 14

Exercise 14B (Pages 316-317)

14

6 Differentiate the following with respect to x.

a e^{6x} \qquad **b** $e^{-x/3}$ \qquad **c** $7e^{2x}$

d $5e^{0.4x}$ \qquad **e** $e^{3x} + 2e^{x}$ \qquad **f** $e^{x}(e^{x} + 1)$

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Exercise 14B (Pages 316-317)

15

7 Find the gradient of the curve with equation $y = e^{3x}$ at the point where

a $x = 2$ \qquad**b** $x = 0$ \qquad**c** $x = -0.5$

8 The function f is defined as $\mathrm{f}(x) = e^{0.2x}$, $x \in \mathbb{R}$. Show that the tangent to the curve at the point $(5, e)$ goes through the origin.

e

e

s

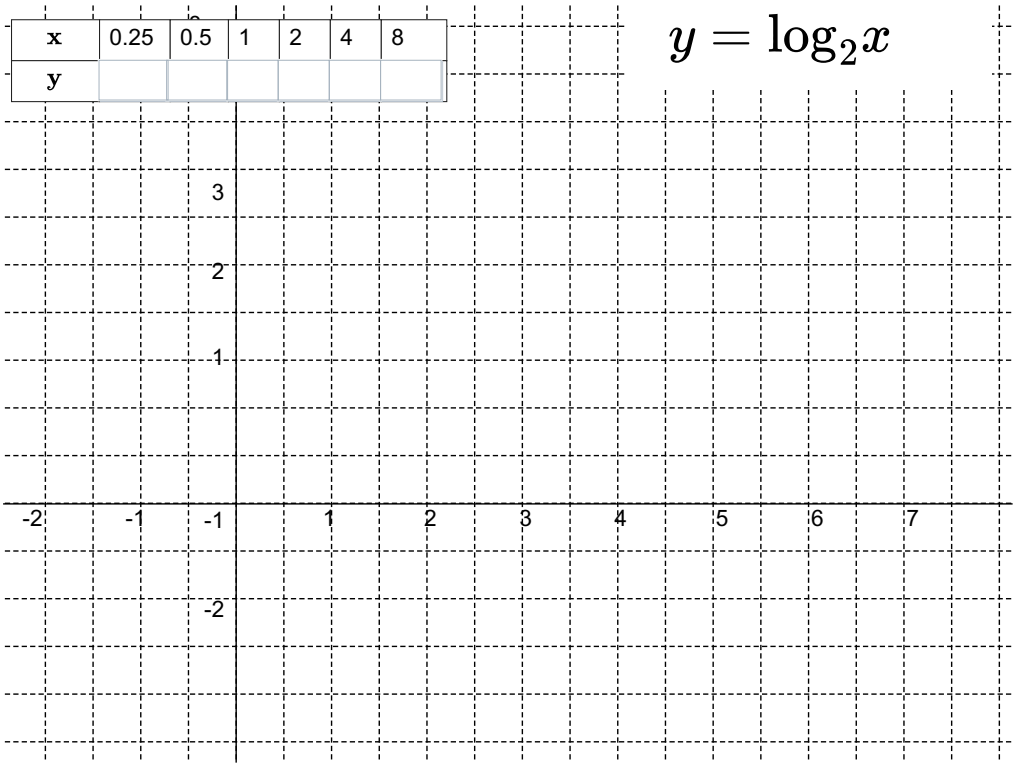
f

Graphs and Transformations

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x	0.25	0.5	1	2	4	8
y						

$y = \log_2 x$



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Natural Logarithms

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We have previously seen that $y = \log_a x$ is the inverse function of $y = a^x$.
We also saw that e^x is "**the**" exponential function.
The inverse of e^x is $\log_e x$, but because of its special importance, it has its own function name!

The inverse of $y = e^x$ is $y = \ln x$

$\ln e^x = \mathbf{x}$
$e^{\ln x} = \mathbf{x}$

Solve $e^x = 5$

Solve $2 \ln x + 1 = 5$